

## 1. PRELIMINARIES

**Definition 1.1** (Multifold sums of powers). *For non-negative integers  $r, n, m$*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

**Definition 1.2** (Central factorials). *For integers  $n, k$*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n \left(n + \frac{k}{2} - 1\right) \left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = n \prod_{j=1}^{k-1} \left(n + \frac{k}{2} - j\right), & \text{if } k > 0. \end{cases}$$

**Proposition 1.3** (Central Newton's formula). *For arbitrary integers  $x$ , and  $t$*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where  $\delta^k f(t)$  denotes central finite difference of  $f(t)$ , defined by

$$\delta^k f(t) = \sum_{j=0}^k (-1)^j \binom{k}{j} f\left(t + \frac{k}{2} - j\right).$$

*Proof.* See [?, p. 462]. □

**Lemma 1.4** (Central falling factorials). *For integers  $n, k$*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n \left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0. \end{cases}$$

where  $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=1}^{k-1} \left(n + \frac{k}{2} - j\right)$  are falling factorials.

**Lemma 1.5** (Binomial form of central factorials). *For integers  $n$  and  $k \geq 1$ ,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1}.$$

*Proof.* We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \left( n + \frac{k}{2} - 1 \right)_{k-1} = \frac{n}{k(k-1)!} \left( n + \frac{k}{2} - 1 \right)_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1},$$

because of the identity in falling factorials  $\frac{(x)_n}{n!} = \binom{x}{n}$ , and Lemma (1.4).  $\square$

**Lemma 1.6** (Halved central factorials). *For integers  $n, k \geq 0$ ,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \binom{n + \frac{k}{2}}{k} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k}.$$

*Proof.* We have  $\binom{a}{k} = \frac{a}{k} \binom{a-1}{k-1}$ . Let  $a = n + \frac{k}{2}$ , then

$$\binom{n + \frac{k}{2}}{k} = \frac{n + \frac{k}{2}}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Thus,

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \binom{n + \frac{k}{2}}{k} - \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \left[ 2 \binom{n + \frac{k}{2}}{k} - \binom{n + \frac{k}{2} - 1}{k-1} \right] = \frac{1}{2} \left[ \binom{n + \frac{k}{2}}{k} + \binom{n + \frac{k}{2} - 1}{k} + \binom{n + \frac{k}{2} - 1}{k-1} - \binom{n + \frac{k}{2} - 1}{k-1} \right].$$

Thus, the statement follows.  $\square$

**Proposition 1.7** (Central Newton's formula for powers). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$n^m = \sum_{k=0}^m \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

**Proposition 1.8** (Halved central factorial powers). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$n^m = \frac{1}{2} \sum_{k=0}^m \left[ \binom{n-t + \frac{k}{2}}{k} + \binom{n-t + \frac{k}{2} - 1}{k} \right] \delta^k t^m$$

**Proposition 1.9** (Generalized hockey-stick identity). *For integers  $a, b$  and  $j$ ,*

$$\sum_{k=a}^b \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

*Proof.* We have,  $\sum_{k=a}^b \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left( \sum_{k=0}^b \binom{k}{j} \right) - \left( \sum_{k=0}^{a-1} \binom{k}{j} \right)$ . By hockey stick identity  $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$  yields,

$$\sum_{k=a}^b \binom{k}{j} = \left( \sum_{k=0}^b \binom{k}{j} \right) - \left( \sum_{k=0}^{a-1} \binom{k}{j} \right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof.  $\square$

**Lemma 1.10** (Centered hockey-stick identity). *For integers  $n \geq 1$ , and arbitrary integers  $t, k, r$*

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

*Proof.* By setting  $j = 1 - t + \frac{k}{2} + r$ , and  $b = n - t + \frac{k}{2} + r$  yields

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k}.$$

By Generalized hockey-stick identity (1.9),

$$\sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Thus, the claim follows.  $\square$

**Proposition 1.11** (Ordinary sums of powers). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \delta^k t^m \sum_{j=1}^n \left[ \binom{j-t+\frac{k}{2}}{k} + \binom{j-t+\frac{k}{2}-1}{k} \right]$$

**Proposition 1.12** (Ordinary sums of powers decomposition). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[ \binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1} \right] \delta^k t^m$$

**Proposition 1.13** (Central factorial numbers sums). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[ \binom{n + \frac{k}{2} + 1}{k+1} - \binom{1 + \frac{k}{2}}{k+1} + \binom{n + \frac{k}{2}}{k+1} - \binom{\frac{k}{2}}{k+1} \right] k! T(m, k),$$

where  $T(n, k)$  denote Central factorial numbers of the second kind (in Riordan's notation), defined by polynomial identity,

$$x^m = \sum_{k=1}^m T(m, k) x^{[k]}.$$

See [?, p. 213], and [?].

**Definition 1.14** (Generalized Central factorial numbers of the second kind). *For integers  $0 \leq k \leq m$ , and an arbitrary integer  $t$ ,*

$$T_t(m, k) = \frac{\delta^k t^m}{k!}.$$

**Proposition 1.15** (Generalized Central factorial sums). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[ \binom{n - t + \frac{k}{2} + 1}{k+1} - \binom{1 - t + \frac{k}{2}}{k+1} + \binom{n - t + \frac{k}{2}}{k+1} - \binom{-t + \frac{k}{2}}{k+1} \right] k! T_t(m, k)$$

**Proposition 1.16** (Double sums of powers decomposition). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\begin{aligned} \Sigma^2 n^m = \frac{1}{2} \sum_{k=0}^m \left[ + \binom{n - t + \frac{k}{2} + 2}{k+2} - \binom{2 - t + \frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{1 - t + \frac{k}{2}}{k+1} \Sigma^1 n^0 \right. \\ \left. + \binom{n - t + \frac{k}{2} + 1}{k+2} - \binom{1 - t + \frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{0 - t + \frac{k}{2}}{k+1} \Sigma^1 n^0 \right] \delta^k t^m \end{aligned}$$

## 2. MATHEMATICA

| Mathematica Function                                       | Validates / Prints              |
|------------------------------------------------------------|---------------------------------|
| <code>MultifoldSumOfPowersRecurrence[r, n, m]</code>       | Computes $\Sigma^r n^m$ , (1.1) |
| <code>CentralFactorial[n, k]</code>                        | Computes $n^{[k]}$ , (1.2)      |
| <code>ValidateCentralFactorialsInTermsOfFalling[10]</code> | Validates Lem. (1.4)            |
| <code>ValidateHalvedCentralBinomials[20]</code>            | Validates Lem. (1.6)            |
| <code>ValidateNewtonsFormulaForPowers[10]</code>           | Validates Prop. (1.7)           |
| <code>ValidateHalvedCentralFactorialPowers[5]</code>       | Validates Prop. (1.8)           |
| <code>ValidateCenteredHockeyStickIdentity[5]</code>        | Validates Lem. (1.10)           |
| <code>ValidateOrdinarySumsOfPowersDecomposition[10]</code> | Validates Lem. (1.12)           |